When Gene Likens and his team investigated why previously fish-rich lakes had died, they found that the culprit was acid rain, caused by emissions of sulfur dioxide and nitrogen oxide from industrial chimneys. As a result, legislation to control the emissions was passed in the US and Europe. After US chemists Frank Rowland and Mario Molina showed that chlorofluorocarbons (CFCs) destroy atmospheric ozone, the use of CFCs was banned worldwide in 1989.

Light pollution, which affects beach-nesting turtles, bats, and migrating birds, has proved harder to control. The International Dark-Sky Association is at the forefront of campaigns for environmentally responsible lighting.

Diminished resources

Garrett Hardin, an American ecologist, warned of the dangers of overpopulation in 1968, when the global population was 3.6 billion. By 2022, it hit 8 billion, and although the growth rate has slowed considerably, the ever-increasing consumption of natural resources has led to depleted stocks of wood, fossil fuels, minerals, and even fish. The collapse of the once bountiful cod fishery off Newfoundland in 1992 highlighted the vulnerability of our food chain to overfishing and led the Canadian government to impose a 32-year moratorium on fishing on the Grand Banks.

Clean and safe drinking water is one of the most fundamental requirements for society, but around 2 billion people do not have access to it. A lethal combination of climate change and population growth in some developing regions threatens to increase this number.